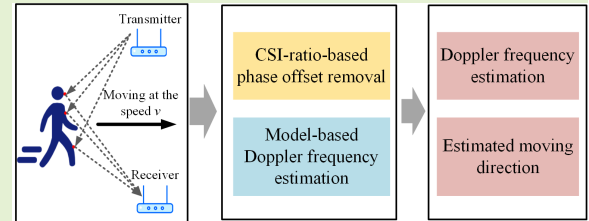


CSI-Ratio-based Doppler Frequency Estimation in Integrated Sensing and Communications

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Abstract—Estimating the Doppler frequency is an important part in sensing moving targets in integrated sensing and communications (ISAC) systems, such as human tracking and activity recognition. However, it can be highly challenging when there is clock asynchronism between the transmitter and receiver, in bistatic setups that are common nowadays. In this paper, we propose three algorithms for Doppler frequency estimation based on the ratio of channel state information (CSI). These algorithms explore different properties of the CSI ratio, including the circle-preserving property of the Mobius transform, the periodicity of CSI ratio, and the difference (or correlation) between segments of CSI-ratio signals. Experimental results demonstrate that the proposed algorithms can estimate Doppler frequency accurately, outperforming the commonly used approach based on cross-antenna cross correlation.

Index Terms—Channel state information (CSI); Clock asynchronism; Cross-antenna CSI ratio; Doppler frequency estimation; Human activity recognition; Integrated Sensing and Communications (ISAC)



I. INTRODUCTION

IN communication-centric integrated sensing and communications (ISAC), communication signals are also used for extracting environmental information and sensing, such as detecting and tracking human targets, and identifying their activities [1], [2]. In a variety of context-aware human sensing tasks, estimating the Doppler frequencies from the received signals is an essential step [3]. For instance, the estimated Doppler frequency can be used to determine respiration pattern and human moving velocity [2]. Passive WiFi radar (PWR) has been designed to estimate Doppler frequency indoors [4], [5]. In PWR, two clock-synchronized receivers collaboratively work, one receiving direct WiFi signals as references and the other receiving echo signals for sensing. Another mainstream of WiFi sensing, which has more joint communications and sensing flavor, employs the by-product of a WiFi communication receiver, namely the channel state information (CSI), which does not require a separate reference receiver [6], [7]. We consider in this paper the latter case.

In such WiFi sensing systems, transmitters (Tx) and sensing receivers (Rx) are generally geometrically separated, like a

bistatic radar system. However, unlike in the bistatic radar, there is typically no common clock between the spatially-separated Tx and (sensing) Rx, which leads to the clock asynchronism issue. This issue causes timing offset (TMO), carrier frequency offset (CFO) and sampling offset in the received signals. For sensing, TMO can result in timing ambiguity and hence ranging ambiguity, and CFO causes Doppler estimation ambiguity and further degrades the accuracy of Doppler frequency estimation; furthermore, clock asynchronism also causes time-varying phase shifts across discontinuous CSI measurements, which hinders the coherently processing of CSI measurements across different timeslots/packets and makes Doppler frequency estimation challenging. Therefore, to obtain accurate Doppler frequency estimation with the use of complex signals instead of the power only, removing the phase offsets induced by clock asynchrony is an essential prerequisite step.

To estimate CSI, a WiFi receiver would generally use the training signals to synchronize with the WiFi transmitter. This level of synchronization, however, is not sufficient for accurate sensing [8]. In particular, the CFO residual in a WiFi receiver can be tens hertz in general. Even it is 10 Hz, the Doppler estimation error can be as large as $1.25 (= 10 \times \frac{3 \times 10^8}{2.4 \times 10^9})$ m/s, for a 2.4 GHz WiFi system. Therefore, the communication synchronization errors must be further addressed for accurate sensing.

To suppress synchronization errors, various solutions can be potentially applied [8]. For a single receiver node with multiple antennas, several techniques may be adopted. One technique is to construct a reference signal from the line-of-sight (LOS) path. Then, the TMO in other reflected echoes

This research was supported partially by the Australian Government through the Australian Research Council's Discovery Projects funding scheme (project DP210101411). (Corresponding author: Yuanhao Cui.)

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can be eliminated by measuring the time-difference of arrival (TDOA) between the reference signal and the echoes. However, this technique is sensitive to the quality of the constructed reference signal. The other commonly used strategy exploits the fact that the clock offsets across multiple antennas at a receiving node are the same due to the shared oscillator clock. Two classes of methods have been developed based on this fact: Cross antenna cross correlation (CACC) and cross-antenna signal (or CSI) ratio (CASR). The CACC method removes phase offset, TMO and CFO by computing the cross-correlation between the signals (or CSI) from different receiving antennas [6], [9]. However, it is challenging to remove the image components generated with the cross-correlation operation and the application of this method is limited due to the requirement for, e.g., a dominating LOS path. In contrast, the CASR method removes offsets by computing the CSI ratio between antennas. The CASR method is simple to implement and can improve the signal-to-interference-plus-noise ratio (SINR) of the received sensing signals, hence resulting in a larger sensing range [7]. Although the potential of CASR has been well demonstrated in respiration pattern detection [7], [10] and straight-line movement [11], it has rarely been studied for more complicated scenarios involving irregular and fast movement. A more detailed review of related work will be provided in Section V.

In this paper, we study Doppler frequency estimation using the CASR method for scenarios with general movement. We first develop a CSI-ratio expression disclosing more insights, using a more general CSI model as a function of delay, Doppler frequency and angle-of-arrival (AoA), instead of the widely used and simpler one only based on signal propagation distance [7], [10], [11]. We then propose three algorithms for estimating Doppler frequencies: Mobius transformation-based, signal difference-based, and periodicity-based ones. These algorithms exploit different features of the CSI ratio in terms of Doppler frequencies and can be applied to scenarios involving general and/or irregular movement. We describe these algorithms by referring to human tracking applications in this paper, but they can be easily adapted to other applications. Using a publically available WiFi CSI dataset *Widar 2.0*, we then validate the efficiency of the proposed Doppler frequency estimation algorithms.

The rest of the paper is organized as follows. In Section II, we present and analyze the CSI ratio based on a general CSI model. Three CSI-ratio-based Doppler frequency estimation algorithms are provided in Section III. In Section IV, experimental results are presented to validate the efficiency of the proposed methods. In Section V, we provide a more detailed review of related work on RF-based human sensing and phase offset removal in bi-static wireless sensing systems. Finally, the conclusion is provided in Section VI.

II. SENSING SIGNAL MODEL

In this section, we first introduce the basic concept of CSI and present the relationship between the Doppler frequency of a moving human target and the change of CSI measurements. Then, we adopt the cross-antenna signal ratio strategy to solve

the clock synchronization problem, and describe the properties of the produced CSI-ratio measurements when being utilized for Doppler frequency estimation.

A. CSI Model

We consider a multiple-input multiple-output (MIMO-OFDM) system with M_T transmit antennas and M_R receive antennas. Let B denote the total signal bandwidth, N be the number of total subcarriers, $f_0 = B/N$ be subcarrier interval, and T_s be the OFDM symbol period.

Consider a planar wave-front signal propagation model. The array steering vector of a uniform linear antenna array (ULA) at the receiver is given by

$$\mathbf{a}(M_R, \alpha) = [1, e^{ju(\alpha)}, \dots, e^{j(M-1)u(\alpha)}]^T, \quad (1)$$

where $u(\alpha) = 2\pi d/\lambda \sin(\alpha)$, λ is the wavelength, d is the antenna spacing, and α is the angle-of-arrival (AoA) of a signal path. Similarly, we can define $\mathbf{a}(M_T, \beta)$ for the transmitter array, where β is the angle of departure (AoD).

The CSI acquired at a WiFi receiver can be imperfect due to random phase noise, TMO and CFO. These nuisances are included in the signal model below. The frequency-domain OFDM channel state matrix $\mathbf{H}(t)$ at the n -th subcarrier at time t can be represented as [1]

$$\mathbf{H}_n(t) = e^{j\phi_n(t)} \sum_{l=1}^L b_l e^{-j2\pi n(\tau_l + \tau_o(t))f_0} e^{j2\pi(f_{D,l} + f_o(t))t} \mathbf{a}(M_R, \beta_l) \mathbf{a}^T(M_T, \beta_l), \quad (2)$$

where b_l is the amplitude of the l -th multipath; τ_l is the propagation delay; $f_{D,l}$ is the Doppler frequency caused by the moving human target; $\phi_n(t)$, $\tau_o(t)$ and $f_o(t)$ are the time-varying phase shift, TMO and CFO induced by clock asynchronism between transmitters and receivers, respectively.

Since our algorithms to be proposed can be similarly applied to each subcarrier, we drop the subscript n hereafter. We also consider the CSI with a single transmitting antenna. For multiple transmitting antennas, the channel matrix can be easily separated for each antenna as the training sequences across transmitting antennas are generally orthogonal. Assume that there is only one dynamic path and L_s ($L_s \geq 1$) static paths, which corresponds to approximating the single mobile human target to be sensed as a point source.

Under the above setup, we rewrite the channel CSI at the m -th receive antenna, $H^m(t)$, as the sum of static paths and dynamic path. Let

$$H_s^m(t) = \sum_{l_s=1}^{L_s} b_{l_s} e^{-j2\pi n \tau_{l_s} f_0} e^{j(m-1)u(\theta_{l_s})}, \quad (3)$$

and

$$H_d^m(t) = b_d e^{-j2\pi n \tau_d f_0} e^{j(m-1)u(\alpha)} \quad (4)$$

represent parts of the static and dynamic paths, respectively, where τ_{l_s} and θ_{l_s} are the TMO and AoA of the l -th static path, respectively; τ_d and α are the TMO and AoA of the dynamic path, respectively, and $u(\theta_{l_s}) \triangleq 2\pi d/\lambda \sin(\theta_{l_s})$. Note that the

AoD related phase term has been absorbed into b_{l_s} and b_d . We can then represent $H^m(t)$ as

$$H^m(t) = e^{j\phi_n(t)} e^{j2\pi(f_o(t)t - n\tau_o(t)f_o)} \cdot (H_s^m(t) + H_d^m(t)e^{j2\pi f_D t}). \quad (5)$$

B. CSI-Ratio Model

The CSI ratio $R(t)$ between the m -th and $(m+1)$ -th receiving antennas can be expressed as

$$R(t) = \frac{H^m(t)}{H^{m+1}(t)} = \frac{H_s^m(t) + H_d^m(t)z(t)}{H_s^{m+1}(t) + H_d^{m+1}(t)z(t)},$$

where

$$z(t) \triangleq e^{2\pi f_D t}. \quad (7)$$

Note that the CSI imperfections including phase noise, TMO and CFO are suppressed in (6), as they are common to all receiving antennas and hence the numerator and denominator therein.

The varying sensing parameters, i.e., Doppler frequency, propagation delay and AoA, are very different due to the movement of a human target. AoA typically changes most slowly, and then delay and Doppler frequency. A human target generally moves less than five meters per second indoors. Thus, we can reasonably assume they are all fixed over a period of tens of milliseconds. Therefore, in such a short time period, the environment illuminated by the transmitted signals can be treated as static. Then, $H_s^m(t)$, $H_s^{m+1}(t)$, $H_d^m(t)$ and $H_d^{m+1}(t)$ are constant, and can be denoted as H_s^m , H_s^{m+1} , H_d^m and H_d^{m+1} , respectively. In this case, $R(t)$ only varies with $z(t)$ and can be rewritten as

$$R(t) = \frac{H_s^m + H_d^m e^{j2\pi f_D t}}{H_s^{m+1} + H_d^{m+1} e^{j2\pi f_D t}} = \frac{H_s^m + H_d^m e^{j2\pi f_D t}}{H_s^{m+1} + H_d^{m+1} e^{j2\pi f_D t}}. \quad (8)$$

Collecting CSI measurements over this period, we can then estimate the Doppler frequency f_D based on (8). Generally, there are some imperfections in the acquired CSI data, such as the residual CFO and TMO. With the proposed CSI-ratio model, these imperfections in CSI data can be well canceled by computing the CSI ratios between one reference antenna and others.

III. PROPOSED DOPPLER FREQUENCY ESTIMATION METHODS

In this section, we propose three Doppler frequency estimation algorithms, by exploiting several different properties based on the CSI-ratio model developed above. These algorithms estimate a Doppler frequency for each segment of CSI-ratio samples, over a time window of tens-of-millisecond.

Firstly, we apply a calibration step to avoid the estimation of a pseudo Doppler frequency when the target is moving at almost zero velocity or is stationary. In this case, the phase of CSI data changes slightly due to the environmental noise. If

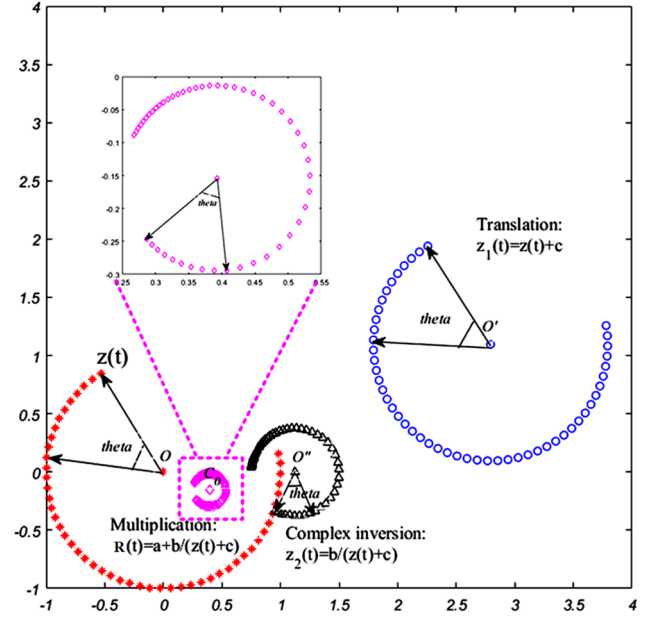


Fig. 1. Illustration of Mobius Transform. With the translation, complex inversion, multiplication operations, $z(t)$ is transformed to $R(t)$, which is the Mobius transform of $z(t)$.

this case is not identified in advance, our proposed algorithms will estimate a pseudo Doppler frequency. To deal with this issue, we calculate the square deviation of CSI amplitude in the time window, and compare the deviation value with a preset threshold. If the deviation is larger than the threshold, we then activate one of the proposed algorithms. Otherwise, the Doppler frequency in this time window is set to 0 Hz.

Next, we describe the three algorithms.

A. Doppler Frequency Estimation based on Mobius Transformation

In the complex plane, with t increasing, $z(t)$ is a unit circle rotating clockwise or anticlockwise, depending on the sign of f_D . When H_s^m , H_s^{m+1} , H_d^m and H_d^{m+1} are invariant and satisfy $H_s^m H_d^m - H_s^{m+1} H_d^{m+1} \neq 0$, $R(t)$ can be treated as the Mobius transformation of $z(t)$ in the complex plane, as illustrated in Fig. 1.

After the translation, complex inversion, and multiplication transform in Fig. 1, we can see that the CSI ratio also changes along a circle in the complex plane, although the speed of variations become non-uniform, due to the inversion transform. When the magnitude of the static component is larger than that of the dynamic one, the CSI ratio $R(t)$ and $z(t)$ rotate in the same direction in the complex plane; otherwise, they rotate in opposite directions. For human sensing applications in real scenarios, the magnitude of the static component H_s^m is generally larger than that of the dynamic component H_d^m [7]. In this case, the rotation direction of $z(t)$ can be inferred based on the direction of $R(t)$.

Furthermore, according to the Mobius transformation, $z(t) \rightarrow R(t)$ is a conformal map, as shown in Fig. 1 [12]. Let $\theta_R(t)$ be the angle of a point at $R(t)$ with respect to its center C_0 . That is, $\theta_R(t) = \angle(R(t) - C_0)$. Then, at some

Algorithm 1 Doppler Frequency f_D Estimation based on Mobius Transform.

Input:

A sequence of CSI-ratio samples $R(t_k)_{k=1}^N = \{R(t_1), R(t_2), \dots, R(t_N)\}$ within Δt , where $\Delta t = t_N - t_1$;

Output:

The estimated f_D within $[t_1, t_N]$;

- 1: The coordinate of the center C_0 of $R(t_k)_{k=1}^N$ is denoted as $[A, B]$.
- 2: Find the values of A and B that make the sum of deviation δ_k^2 in Eq. (11) reaches the minimum value;
- 3: The new CSI-ratio samples $R_s(t_k)_{k=1}^N$ are calculated as $R(t_k)_{k=1}^N - C_0$;
- 4: The angle of $R_s(t_k)_{k=1}^N$ are denoted as $\theta_R(t_1), \theta_R(t_2), \dots$, and $\theta_R(t_N)$, and the magnitude are $a_R(t_1), a_R(t_2), \dots$, and $a_R(t_N)$;
- 5: Estimate the value $\beta_1 = 2\pi T_s f_D$ with the angle and the magnitude of $R_s(t_k)_{k=1}^N$, by using the weighted linear fitting methods in Eq. (14);
- 6: $f_D = \beta_1 / 2\pi T_s$;
- 7: **return** Estimated f_D within the time period $[t_1, t_N]$;

relatively larger time interval Δt , the angle variation $\Delta(\theta_z)$ of $z(t)$ and $\Delta(\theta_R)$ of $R(t)$ is approximately equal, i.e.,

$$\Delta(\theta_R) \approx \Delta(\theta_z). \quad (9)$$

When Δt is a few milliseconds, the Doppler frequency $f_D = \frac{v_D}{c} f_c$ of the human target can be regarded as invariant, where v_D is the relative radical speed of the human target, f_c is the carrier frequency, and c is the speed of the light. Therefore,

$$\Delta(\theta_R) \approx \Delta(\theta_z) = 2\pi f_D \Delta t. \quad (10)$$

To obtain $\Delta(\theta_R)$, we need to estimate the center C_0 of the CSI-ratio samples $R(t_k)_{k=1}^N = \{R(t_1), R(t_2), \dots, R(t_N)\}$ within Δt , where $\Delta t = t_N - t_1$. We adopt the least square method to estimate C_0 .

Proposition 3.1: In the complex plane, let the coordinate of C_0 be (A, B) and the radius of the CSI-ratio circle be r , the coordinate of C_0 can be obtained by

$$\min_{A, B, r} \sum_{k=1}^N \delta_k^2 = \min_{A, B, r} \sum_{k=1}^N ((x_k - A)^2 + (y_k - B)^2 - r^2)^2, \quad (11)$$

where (x_k, y_k) is the coordinate of $R(t_k)$ in the complex plane.

Proof: Please refer to the Appendix.

Note that not a whole circle samples are needed to get the estimate of C_0 , and quite often a segment of arc is sufficient. Then, the coordinate of C_0 can be estimated as given in Eq. (35). With the estimated C_0 , the coordinate of $R(t_k)_{k=1}^N$ with respect to C_0 can be calculated. Then, the new CSI-ratio samples $R_s(t_k)_{k=1}^N$ is given by

$$R_s(t_k)_{k=1}^N = R(t_k)_{k=1}^N - C_0, \quad (12)$$

Algorithm 2 Doppler Frequency f_D Estimation based on Periodicity of CSI Ratio.

Input:

The angle of a sequence of CSI-ratio samples $\{\gamma(t_1), \gamma(t_2), \dots, \gamma(t_N)\}$ within the time period $[t_1, t_N]$;

The minimum sample size S_{min} in one cycle;

The maximum sample size S_{max} in one cycle;

Step I ;

Output:

The estimated f_D within $[t_1, t_N]$;

- 1: Initialize $k = 1$;
- 2: Initialize an empty list F_list ;
- 3: **while** $k \leq N - S_{max}$
- 4: The angle sample $\gamma(t_k)$ is denoted as γ_{base} ;
- 5: $j = k + 1$;
- 6: **while** $j - k \leq S_{max}$
- 7: **if** $(\gamma(t_j) - \gamma(t_k))(\gamma(t_{j+1}) - \gamma(t_k)) \leq 0$ and $S_{min} \leq (j - i)$
- 8: $S = j - k$;
- 9: **break**;
- 10: **else**
- 11: $S = S_{max}$;
- 12: **end**
- 13: $j = j + 1$;
- 14: **end**
- 15: $f_D = \frac{1}{S T_s}$;
- 16: Add f_D to F_list ;
- 17: $k = k + I$;
- 18: **end**
- 19: Average the f_D values in F_list , and f_D in $[t_1, t_N]$ is estimated to be the averaged value.
- 20: **return** Estimated f_D within the time period $[t_1, t_N]$;

which equivalently shifts the center of $R(t)$ to the origin of the complex plane.

Therefore, the angle $\theta_R(t_k)_{k=1}^N = \{\theta_R(t_1), \theta_R(t_2), \dots, \theta_R(t_N)\}$ and the magnitude (i.e., absolute value of the amplitude) $a_R(t_k)_{k=1}^N = \{a_R(t_1), a_R(t_2), \dots, a_R(t_N)\}$ of the new CSI-ratio samples $R_s(t_k)_{k=1}^N$ with respect to C_0 can be obtained. Then, based on (10), when f_D is invariant, $(t_k, \theta_R(t_k))_{k=1}^N = \{(t_1, \theta_R(t_1)), (t_2, \theta_R(t_2)), \dots, (t_N, \theta_R(t_N))\}$ form a linear mapping, i.e.,

$$\theta_R(t_k) = \beta_1 t_k + \beta_2, \quad (13)$$

where β_1 is equal to $2\pi f_D T_s$.

To get the estimate for β_1 , we adopt a weighted linear fitting method

$$\min_{\beta_1, \beta_2} Q(\beta_1, \beta_2) = \min_{\beta_1, \beta_2} \sum_{k=1}^N w_k (\theta_R(t_k) - (\beta_1 t_k + \beta_2))^2, \quad (14)$$

where we use the magnitude $a_R(t_k)$ as the weight w_k . The weights can reduce the impact of phase errors of small ratio. Then, the estimated f_D can be calculated as $\beta_1 / (2\pi T_s)$.

The main steps of estimating f_D with the Mobius-Transformation-based algorithm are illustrated in Algorithm 1. Note that this method can directly estimate the sign of the

Doppler frequency, and is the only one with this capability in the proposed three algorithms in this paper.

Algorithm 3 Doppler Frequency f_D Estimation based on Signal Difference.

Input:

A sequence of CSI-ratio samples $\{R(n)\}_{n=1}^N$ within the time period $[t_1, t_N]$; The length k_0 of the reference sample dataset $\{R(1), R(2), \dots, R(k_0)\}$; Pseudo-periodicity n_0 ;

Output:

The estimated f_D within $[t_1, t_N]$;

- 1: Initialize an empty list D_list ;
- 2: **while** $n_0 \leq n \leq N$
- 3: Initialize $\Delta_s = 0$;
- 4: **while** $k_0 \leq k \leq k_0 + K - 1 \leq N$
- 5: $\Delta_R^2(k, n) = |R(k+n) - R(k)|^2$
- 6: $\Delta_s = \Delta_s + \Delta_R^2(k, n)$;
- 7: **end**
- 8: Add Δ_s to D_list ;
- 9: **end**
- 10: Find the value n^* from D_list that satisfies $n^* = \arg_n \min \frac{1}{K} \sum_{k=k_0}^{k_0+K-1} \Delta_R^2(k, n)$;
- 11: $f_D = \frac{1}{n^* T_s}$;
- 12: **return** Estimated f_D within the time period $[t_1, t_N]$;

B. Doppler Frequency Estimation based on Periodicity of CSI Ratio

Since $z(t) \rightarrow R(t)$ is a conformal map, $z(t)$ and $R(t)$ have the same periodicity. Therefore, when the radian of $z(t)$ changes by 2π , $R(t)$ goes exactly one period. In this case, we can estimate f_D based on the periodicity of $R(t)$. Specifically, for a sequence of CSI-ratio samples $R(t) = \{R(t_1), R(t_2), \dots, R(t_N)\}$, denote their angles relative to the point $O=(0, 0)$ on the complex plane as $\gamma(t) = \{\gamma(t_1), \gamma(t_2), \dots, \gamma(t_N)\}$. Then, if there are two angles separated by a reasonable amount of time are similar, we can infer that $R(t)$ goes through a cycle within this period Δt . Then, f_D can be estimated as

$$f_D = \frac{2\pi}{2\pi\Delta t} = \frac{1}{\Delta t}. \quad (15)$$

The detailed f_D estimation process based on the periodicity of CSI ratio is presented in Algorithm 2.

C. Doppler Frequency Estimation based on Signal Difference/Correlation

Since H_s^m, H_s^{m+1}, H_d^m and H_d^{m+1} are time-invariant, two segment of signals of $R(t)$ separated by n^*T_s ($f_d n^*T_s \approx 1$) will exhibit high correlation. In this subsection, we exploit such signal difference (or correlation) to estimate f_D .

Rewrite (8) as

$$R(t) = a + \frac{b}{c + z(t)}, \quad (16)$$

where $b \triangleq (H_d^m H_s^{m+1} - H_s^m H_d^{m+1})/H_s^{m+1}$, $a \triangleq H_s^m/H_s^{m+1}$, and $c \triangleq H_d^{m+1}/H_s^{m+1}$. Here a , b , and c are

also time-invariant. With $t_k = kT_s$, $R(t_k)$ can be simplified as $R(k) = a + b/(c + z(k))$.

Then, the square of the difference $\Delta_R^2(n)$ between two CSI-ratio samples $R(k)$ and $R(k+n)$ can be represented as

$$\begin{aligned} \Delta_R^2(k, n) &= |R(k+n) - R(k)|^2 \\ &= |b|^2 \cdot \left| \frac{e^{j(k+n)v} - e^{jkv}}{(c + e^{jkv})(c + e^{j(k+n)v})} \right|^2, \end{aligned} \quad (17)$$

where $v \triangleq 2\pi f_D T_s$. From (17), we can see that when $e^{jn2\pi f_D T_s} = 1$, i.e., $n f_D T_s = 1$, $\Delta_R(k, n)$ is 0. This is the ideal case when a sequence of signals fully repeat themselves and noise is absent. In practice, it may not be exactly zero. To exploit the averaging effect, we instead compute the difference (or correlation) between two segment of signals spaced at different distances, and then look for the minimum (or maximum correlation) of the output. Using the difference, this can be presented as

$$n^* = \arg_n \min \frac{1}{K} \sum_{k=k_0}^{k_0+K-1} \Delta_R^2(k, n), n = n_0, \dots, N; \quad (18)$$

$$f_D = 1/(n^* T_s), \quad (19)$$

where $0 \leq K \leq N$. Note that two pieces of $R(k)$ slightly separated may also exhibit high correlation as the phase differences caused by the Doppler frequency could be small. Therefore, we shall either start from a relatively large n_0 and then look for the first local minimum, or start from $n_0 = 1$ but look for the first “bottom” of the difference power (where curves go down first and then up). The latter is easier to implement without the trouble of determining the value of n_0 and is generally more reliable.

The detailed implementation of this method is illustrated in Algorithm 3. Before ending the section, we analyze and compare the computational complexity of the three algorithms. First, as shown in the pseudocode of Algorithms 1, since each statement from Line 1 through Line 7 is executed once for each estimation of Doppler frequency, the complexity of the Mobius-based method is $\mathcal{O}(7) = \mathcal{O}(1)$. In a similar way, based on Algorithm 2, the complexity of periodicity-based method can be calculated as

$$\mathcal{O}(3 + 5(N - S_{max}) + (N - S_{max}) \times S_{max} \times 3) = \mathcal{O}(N).$$

According to Algorithm 3, the complexity of difference-based can be given by

$$\mathcal{O}(3 + 2N + 2N \times N) = \mathcal{O}(N^2).$$

Clearly, the computational complexity of Algorithm 1 is lowest, while the complexity of Algorithm 3 is the highest. The higher complexity, however, is the cost for better estimation accuracy, which will be seen from the results given next.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

To verify the effectiveness of the proposed methods, we conduct experiments by employing the public WiFi CSI dataset *Widar 2.0* [13]. The system uses one single transmitting antenna and three receiving antennas. The Intel 5300 WiFi card

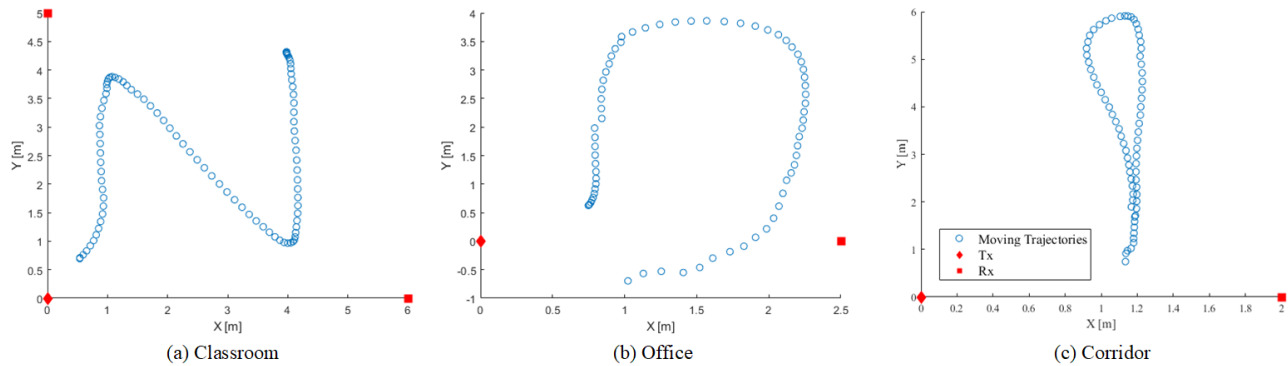


Fig. 2. Three trajectories of human movements in three scenarios.

is used for CSI collection. The human motion channel CSI were collected in three environments: classroom, office and corridor. Several typical trajectories of human movements are shown in Fig. 2. When implementing the proposed algorithms, we use a sequence of 128 consecutive CSI results for each Doppler frequency estimation. The sample frequency of the WiFi-sensing system is 2 kHz [13]. Thus, each Doppler frequency estimation is obtained in 64 ms.

We estimate the Doppler frequency of a human target moving in the three environments, using the three proposed algorithms, and compare the results with those obtained by the CACC method in *IndoTrack* [6]. The estimation results are illustrated in Fig. 3 - Fig. 5 for classroom, office and corridor, respectively. Specifically, for the CACC method, we perform STFT on the CSI data after the antenna selection and the conjugate multiplication operations, and transform the data into a 2-dimensional time-Doppler frequency map, as shown in Fig. 3(a) - Fig. 5(a). Furthermore, at every time point, we select the frequency component that has the maximum power value, and treat these frequency components as the estimated Doppler frequencies, as shown in the black marks in Fig. 3(b) - Fig. 5(b). The sign of the Doppler frequency estimate in the Mobius-based algorithm is used as the sign in the other two algorithms.

From the estimated Doppler frequency results, we can see that the proposed three algorithms generally perform well, demonstrating much smaller fluctuations, compared to the CACC method in [6]. It is because that the image components are not effectively removed in [6], and cause large deviations in the estimates.

Overall, the signal difference-based algorithm achieves the best performance in the three scenarios, although it is incapable of estimating the sign of target Doppler frequency. The calibrated results of the difference-based method is also shown in Fig. 6. Comparing with the results in Fig. 3(b) and 4(b), we can see that with calibration, the performance is notably improved when the target moves at a low speed.

The performance of the Mobius-based algorithm is inferior to that of the difference-based one. Despite of this, the Mobius-based method is capable of estimating both the value and sign of a Doppler frequency. This is a very important feature.

Compared with the other two proposed algorithms, the

periodicity-based method appears to achieve less accurate estimates. This is probably because unlike the other two algorithms, it does not explore the diversity effect of exploring multiple samples in the estimation and hence is susceptible to the noise. The advantage of this approach is that it is easy and simple to implement, and can be used for a rough or initial estimation of target Doppler frequency.

V. RELATED WORK

A. Learning-based and Model-based RF-based Sensing

RF-based human sensing has gained increasing popularity in recent years. There are two categories of human sensing approaches, which are learning-based and model-based, respectively. The learning-based method tries to learn the mapping between the input data and the output label by using a huge amount of training samples. As a typical and effective learning-based method, deep learning has been widely used for RF-based human sensing, and has made significant progress. For instance, Chen et al. [14] proposed an attention-based bi-directional long short-term memory (BLSTM) model with WiFi CSI data for human activity recognition. Alazrai et al. [15] transformed raw WiFi CSI data into a series of 3D "time- frequency- spatial" images and proposed a convolutional neural network (CNN) to recognize human-to-human interactions. Shi et al. [15] proposed a deep learning based activity recognition method by using only one-shot training samples. However, deep learning (DL) method requires a huge amount of training samples to learn the data distribution. Especially, the performance of deep learning approaches generally improves with the labelled training samples increasing, which leads to the burden of data segmentation and annotation. Furthermore, it is difficult for deep learning algorithms to characterize the physical characteristics of electromagnetic echo signals, because traditional signal processing mainly relies on statistical modelling technology. As a result, the reliability and interpretability of DL models may be poor.

Compared with learning-based approaches, the model-based approaches can model the relationship between CSI dynamics and human movements mathematically, and can achieve better performance when used for fine-grained applications [3]. Furthermore, with solid theoretical derivation, model-based methods are more interpretable and reliable. There have

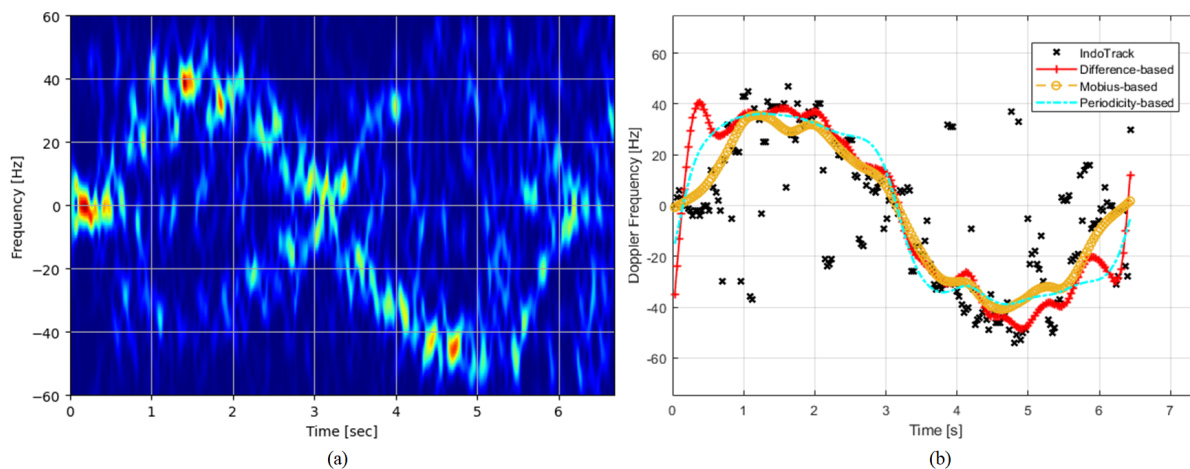


Fig. 3. Doppler frequency estimation results in the classroom scenarios. (a) The estimation results with the *IndoTrack* method [6]. (b) The estimation results with the three proposed methods.

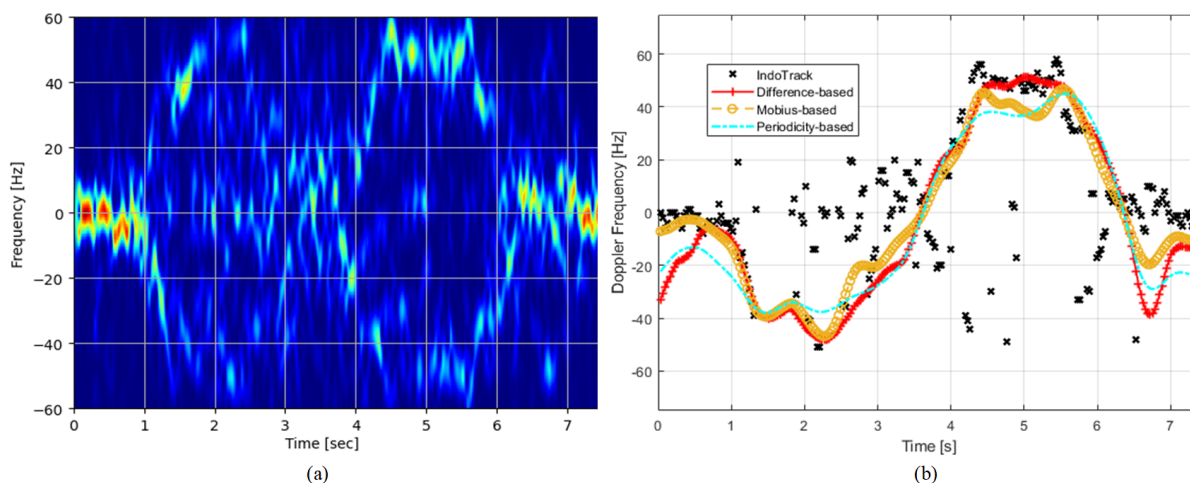


Fig. 4. Doppler frequency estimation results in the office scenarios. (a) The estimation results with the *IndoTrack* method [6]. (b) The estimation results with the three proposed methods.

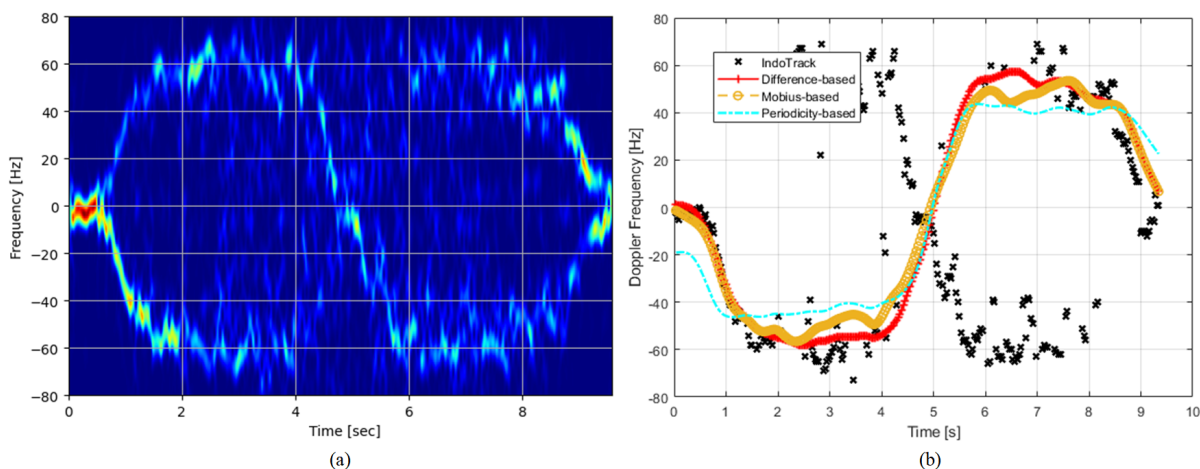


Fig. 5. Doppler frequency estimation results in the corridor scenarios. (a) The estimation results with the *IndoTrack* method [6]. (b) The estimation results with the three proposed methods.

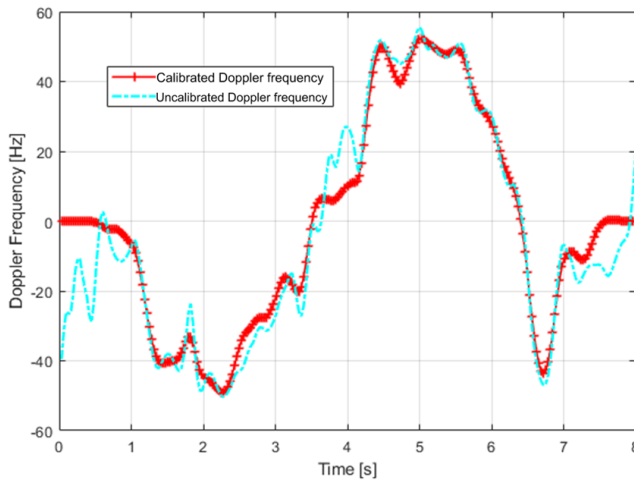


Fig. 6. The estimated Doppler frequency in the office scenario with the calibrated difference-based approach.

been a series of CSI physical models proposed, such as the AoA model, the Fresnel model, and the CSI-ratio model. Specifically, the AoA model [16] inferred human activities by estimating the angle of incident signals at the receiving node, and is generally used for human target localization and tracking tasks. Zhang et al. [17] proposed the Fresnel model, which divides the space between and around the transceiver into several concentric prolate ellipsoidal regions, i.e., Fresnel regions. In this case, the amplitude and phase of CSI data can be correlated with human movements. Especially, once a human target moves across different Fresnel regions, the amplitude and phase of CSI will change. Based on the Fresnel model, some variants have been proposed. For instance, the Fresnel penetration model [18] mathematically correlates the Fresnel phase difference between two different subcarriers with the target's location. Furthermore, the CSI-ratio model [7] was proposed to quantify the relationship between human movements and the variations of CSI data from two receiving antennas. In this paper, we adopt the CSI-ratio strategy, and propose the Doppler frequency estimation methods by correlating the CSI-ratio model with human moving speeds.

B. Phase Offset Removal in Wireless Sensing System

In commodity communication systems, e.g., Wi-Fi systems, the receivers are not tightly synchronized with the transmitter in terms of carrier frequency and time, resulting in random phase offsets in received CSI measurements. To ensure the accuracy of Doppler shift estimation, the random phase offsets need to be handled in advance.

To remove the phase offsets at the receiving node, several solutions have been proposed. In addition to the usual method of constructing a reference signal, CACC and CASR are two typical solutions. Specifically, the CACC approach removes the random phase offsets by computing the cross-product between signals from multiple receiving antennas. However, the conjugate multiplication operation introduces extra L terms. To remove extra terms and estimate f_D , it is widely assumed that there exists a dominating line-of-sight (LOS) with a much

larger magnitude than non-line-of-sight (NLOS) paths. In this case, the results of the conjugate multiplication operation can be divided into three parts: the product of static paths of two antennas, the product of the dynamic paths, and the product of the static paths of one antenna and the dynamic paths of another antenna. Since the Doppler shift information is contained in the third part, the first two parts need to be removed with a bandpass filter (BPF). Furthermore, image components in the third part could degrade the performance of Doppler shift estimation and needs to be suppressed. To deal with this, a power adjustment strategy [6] was proposed to increase the power of the term containing the correct Doppler velocity information, and make Doppler frequencies identified in the spectrogram. At the same time, other alternative schemes such as mirrored MUSIC algorithm [9] were also proposed.

Differently, CASR removes the random offsets of multiple receiving antennas via the ratio between CSI signals. Compared with the CACC method, CASR produces an expression where only one term in the denominator contains the parameters to be estimated. Experiments demonstrated that the CASR method can effectively cancel out the noise and improve the SINR of received sensing signals, enlarging the sensing ranges [11]. Based on CASR, a human respiration monitoring method was proposed [7], which quantifies the relationship between the phase variations of CSI ratio and the chest movement. Furthermore, a respiration monitoring method of multiple persons was proposed [10]. In this method, multi-person respiration sensing was modelled as a blind source separation (BSS) problem when the reflected signals of multiple persons are linearly mixed. Then, the independent component analysis (ICA) algorithm was adopted to separate the mixed signals for further single-person respiration monitoring. In this paper, we use the principle of CASR, and propose three algorithms to estimate the Doppler frequency of human targets.

VI. CONCLUSION

In this paper, we have proposed three Doppler frequency estimation algorithms based on the CSI ratio across antennas, for applications involving sensing of moving targets such as human activity recognition and mobile tracking. Among them, the signal difference-based algorithm has the best and the most robust performance for Doppler frequency estimation, while the periodicity-based algorithm is the easiest one to implement. However, these two algorithms are not able to estimate the sign of Doppler frequency. In contrast, the Mobius-based method has the ability of estimating both the sign and value of Doppler frequencies. As a result, the best solution may be combining the strengths of the three algorithms. The current work may also be the cornerstone for estimating other moving parameters based on the CSI-ratio model, e.g., time delay and AoA of human targets. Another important future work is to remove the sign ambiguity of Algorithms 2 and 3, where the range and angle parameters can be jointly exploited to infer the sign of Doppler frequency.

APPENDIX

Based on the Mobius transform, when there is no noise, the CSI-ratio samples $R(t_k)_{k=1}^n = \{R(t_1), R(t_2), \dots, R(t_n)\}$ are

on a circle of the complex plane. Let the coordinates of the center be (A, B) and the radius be r .

Then, the distance d_k between the k th CSI-ratio sample $R(t_k) = (x_k, y_k)$ ($k = 1, 2, \dots, n$) and the center C_0 can be denoted as

$$d_k^2 = (x_k - A)^2 + (y_k - B)^2. \quad (20)$$

The square deviation δ_k between d_k and r can be calculated as

$$\delta_k = d_k^2 - r^2 = (x_k - A)^2 + (y_k - B)^2 - r^2, \quad (21)$$

which should be zero for all samples on the circle.

To accurately estimate C_0 , we seek the value of A, B , and r that can minimize the sum of deviation $Q(A, B, r)$. The derivatives can be computed as

$$\begin{aligned} Q(A, B, r) &= \sum_{k=1}^n \delta_k^2 \\ &= \sum_{k=1}^n ((x_k - A)^2 + (y_k - B)^2 - r^2)^2 \\ &= \sum_{k=1}^n (x_k^2 - 2Ax_k + A^2 + y_k^2 - 2By_k + B^2 - r^2)^2. \end{aligned} \quad (22)$$

Let a, b and c equal to $-2A, -2B$, and $A^2 + B^2 - r^2$, respectively, then

$$Q(A, B, r) = Q(a, b, c) = \sum_{k=1}^n (x_k^2 + y_k^2 + ax_k + by_k + c)^2. \quad (23)$$

It can be seen from Eq. (23), $Q(a, b, c)$ reaches its minimum when the partial derivatives of this function with respect to a, b and c are all equal to 0, which are given by

$$\frac{\partial Q(a, b, c)}{\partial A} = \sum_{k=1}^n 2(x_k^2 + y_k^2 + ax_k + by_k + c)x_k = 0, \quad (24)$$

$$\frac{\partial Q(a, b, c)}{\partial B} = \sum_{k=1}^n 2(x_k^2 + y_k^2 + ax_k + by_k + c)y_k = 0, \quad (25)$$

and

$$\frac{\partial Q(a, b, c)}{\partial r} = \sum_{k=1}^n 2(x_k^2 + y_k^2 + ax_k + by_k + c) = 0 \quad (26)$$

We construct $n \times$ Eq. (24) - $\sum_{k=1}^n x_k \times$ Eq. (26) to cancel out c , and obtain

$$\begin{aligned} &(n \sum_{k=1}^n x_k^2 - \sum_{k=1}^n x_k \sum_{k=1}^n x_k)a + (n \sum_{k=1}^n x_k y_k - \sum_{k=1}^n x_k \sum_{k=1}^n y_k)b \\ &+ n \sum_{k=1}^n x_k^3 + n \sum_{k=1}^n x_k y_k^2 - \sum_{k=1}^n (x_k^2 + y_k^2) \sum_{k=1}^n x_k = 0. \end{aligned} \quad (27)$$

Similarly, computing $n \times$ Eq. (25) - $\sum_{k=1}^n y_k \times$ Eq. (26) leads to

$$\begin{aligned} &(n \sum_{k=1}^n x_k y_k - \sum_{k=1}^n x_k \sum_{k=1}^n y_k)a + (n \sum_{k=1}^n y_k^2 - \sum_{k=1}^n y_k \sum_{k=1}^n y_k)b \\ &+ n \sum_{k=1}^n x_k^2 y_k + n \sum_{k=1}^n y_k^3 - \sum_{k=1}^n (x_k^2 + y_k^2) \sum_{k=1}^n y_k = 0. \end{aligned} \quad (28)$$

Rewrite Eq. (27) and (28) as

$$\zeta_1 a + \zeta_2 b + \zeta_3 = 0, \quad (29)$$

and

$$\zeta_2 a + \zeta_4 b + \zeta_5 = 0, \quad (30)$$

where

$$\begin{aligned} \zeta_1 &\triangleq n \sum_{k=1}^n x_k^2 - \sum_{k=1}^n x_k \sum_{k=1}^n y_k; \\ \zeta_2 &\triangleq n \sum_{k=1}^n x_k y_k - \sum_{k=1}^n x_k \sum_{k=1}^n y_k; \\ \zeta_3 &\triangleq n \sum_{k=1}^n x_k^3 + n \sum_{k=1}^n x_k y_k^2 - \sum_{k=1}^n (x_k^2 + y_k^2) \sum_{k=1}^n x_k; \\ \zeta_4 &\triangleq n \sum_{k=1}^n y_k^2 - \sum_{k=1}^n y_k y_k; \\ \zeta_5 &\triangleq n \sum_{k=1}^n x_k^2 y_k + n \sum_{k=1}^n y_k^3 - \sum_{k=1}^n (x_k^2 + y_k^2) \sum_{k=1}^n y_k. \end{aligned} \quad (31)$$

Based on (29) and (30), a and b can be computed as

$$a = \frac{\zeta_5 \zeta_2 - \zeta_3 \zeta_4}{\zeta_1 \zeta_4 - \zeta_2^2}, \quad (32)$$

$$b = \frac{\zeta_5 \zeta_1 - \zeta_3 \zeta_2}{\zeta_2^2 - \zeta_4 \zeta_1}. \quad (33)$$

Then,

$$c = -\frac{\sum_{k=1}^n (x_k^2 + y_k^2) + a \sum_{k=1}^n x_k + b \sum_{k=1}^n y_k}{n} \quad (34)$$

Finally, the coordinate $[A, B]$ of C_0 and the radius r can be estimated as

$$\begin{aligned} A &= -a/2, \\ B &= -b/2, \\ r &= \frac{1}{2} \sqrt{a^2 + b^2 - 4c}. \end{aligned} \quad (35)$$

REFERENCES

- [1] J. A. Zhang, F. Liu, C. Masouros, R. W. Heath, Z. Feng, L. Zheng, and A. Petropulu, "An overview of signal processing techniques for joint communication and radar sensing," *IEEE Journal of Selected Topics in Signal Processing*, vol. 15, no. 6, pp. 1295–1315, 2021.
- [2] Y. Ma, G. Zhou, and S. Wang, "Wifi sensing with channel state information: A survey," *ACM Computing Surveys (CSUR)*, vol. 52, no. 3, pp. 1–36, 2019.
- [3] Z. Wang, Z. Huang, C. Zhang, W. Dou, Y. Guo, and D. Chen, "CSI-based human sensing using model-based approaches: a survey," *Journal of Computational Design and Engineering*, vol. 8, no. 2, pp. 510–523, 02 2021. [Online]. Available: <https://doi.org/10.1093/jcde/qwab003>
- [4] Q. Chen, B. Tan, K. Chetty, and K. Woodbridge, "Activity recognition based on micro-doppler signature with in-home wi-fi," in *2016 IEEE 18th International Conference on e-Health Networking, Applications and Services (Healthcom)*, 2016, pp. 1–6.
- [5] C. Tang, W. Li, S. Vishwakarma, F. Shi, S. Julier, and K. Chetty, "Mdpote: Human skeletal motion reconstruction using wifi micro-doppler signatures," *arXiv preprint arXiv:2201.04212*, 2022.
- [6] X. Li, D. Zhang, Q. Lv, J. Xiong, S. Li, Y. Zhang, and H. Mei, "Indotrack: Device-free indoor human tracking with commodity wi-fi," *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.*, vol. 1, no. 3, sep 2017. [Online]. Available: <https://doi.org/10.1145/3130940>
- [7] Y. Zeng, D. Wu, J. Xiong, E. Yi, R. Gao, and D. Zhang, "Farsense: Pushing the range limit of wifi-based respiration sensing with csi ratio of two antennas," *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.*, vol. 3, no. 3, sep 2019. [Online]. Available: <https://doi.org/10.1145/3351279>
- [8] J. A. Zhang, K. Wu, X. Huang, Y. J. Guo, D. Zhang, and R. W. Heath, "Integration of radar sensing into communications with asynchronous transceivers," *IEEE Communications Magazine*, pp. 1–7, 2022.
- [9] Z. Ni, J. A. Zhang, X. Huang, K. Yang, and J. Yuan, "Uplink sensing in perceptive mobile networks with asynchronous transceivers," *IEEE Transactions on Signal Processing*, vol. 69, pp. 1287–1300, 2021.
- [10] Y. Zeng, D. Wu, J. Xiong, J. Liu, Z. Liu, and D. Zhang, "Multisense: Enabling multi-person respiration sensing with commodity wifi," *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.*, vol. 4, no. 3, sep 2020. [Online]. Available: <https://doi.org/10.1145/3411816>
- [11] F. Zhang, Z. Chang, K. Niu, J. Xiong, B. Jin, Q. Lv, and D. Zhang, "Exploring LoRa for long-range through-wall sensing," *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.*, vol. 4, no. 2, jun 2020. [Online]. Available: <https://doi.org/10.1145/3397326>
- [12] J. Olsen, "The geometry of möbius transformations," *Rochester: University of Rochester*, 2010.
- [13] K. Qian, C. Wu, Y. Zhang, G. Zhang, Z. Yang, and Y. Liu, "Widar2. 0: Passive human tracking with a single wi-fi link," in *Proceedings of the 16th Annual International Conference on Mobile Systems, Applications, and Services*, 2018, pp. 350–361.
- [14] Z. Chen, L. Zhang, C. Jiang, Z. Cao, and W. Cui, "Wifi csi based passive human activity recognition using attention based blstm," *IEEE Transactions on Mobile Computing*, vol. 18, no. 11, pp. 2714–2724, 2019.
- [15] Z. Shi, J. A. Zhang, R. Y. Xu, and Q. Cheng, "Environment-robust device-free human activity recognition with channel-state-information enhancement and one-shot learning," *IEEE Transactions on Mobile Computing*, vol. 21, no. 2, pp. 540–554, 2022.
- [16] L. Chen, I. Ahriz, and D. Le Ruyet, "Aoa-aware probabilistic indoor location fingerprinting using channel state information," *IEEE Internet of Things Journal*, vol. 7, no. 11, pp. 10 868–10 883, 2020.
- [17] D. Zhang, H. Wang, and D. Wu, "Toward centimeter-scale human activity sensing with wi-fi signals," *Computer*, vol. 50, no. 1, pp. 48–57, 2017.
- [18] H. Wang, D. Zhang, K. Niu, Q. Lv, Y. Liu, D. Wu, R. Gao, and B. Xie, "Mfdl: A multicarrier fresnel penetration model based device-free localization system leveraging commodity wi-fi cards," 2017. [Online]. Available: <https://arxiv.org/abs/1707.07514>